



PROJECT PRESENTATION

Automatic Production Line Analytics

Version 1.0

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SOLUTION ARCHITECTURE

Automatic Production Line Analytics

1. DATA SOURCES



ARTICLE BASE
Products description
and details



LINE PARAMETERS
Equipment description
and targets



PRODUCTION DATA
Raw production
records



PLANNING DATA
Production plans



QUALITY DATA
Quality and
scrap records

2. SQL SERVER TRANSFORMATION



SQL Server

- Data cleansing
- Business Logic
- Aggregations
- SQL views layer

3. ANALYTICAL MODEL



STAR SCHEMA MODEL

Fact Table: vw_OEEAutoLine
Dimension Tables: Date, Line, Shift

4. DAX CALCULATION LAYER



DAX MEASURES

KPIs, Time Intelligence,
Business Calculations

5. REPORTING AND VISUALIZATION



Power BI

- Interactive Dashboards
- KPIs & Analytics
- Drill-down Analysis
- Data-driven decisions

BUSINESS VALUE

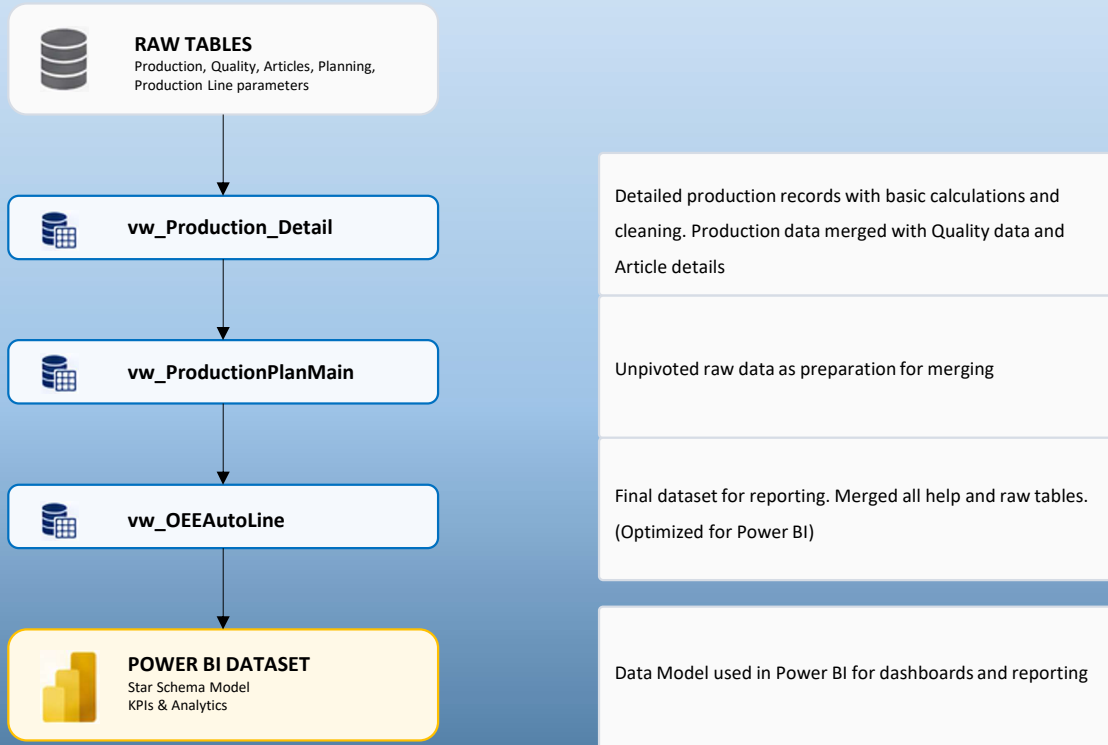


- ✓ Improve Efficiency
- ✓ Data-Driven Decisions
- ✓ Reduce Losses
- ✓ Increase visibility
- ✓ Continuous Improvement
- ✓ Fast Reactivity



SQL DATA PIPELINE

From Raw Data to Power BI Dataset



KEY BENEFITS



**Optimized
Performance**



**Data
Consistency**



**Reusable
Business Logic**



**Scalable
Architecture**

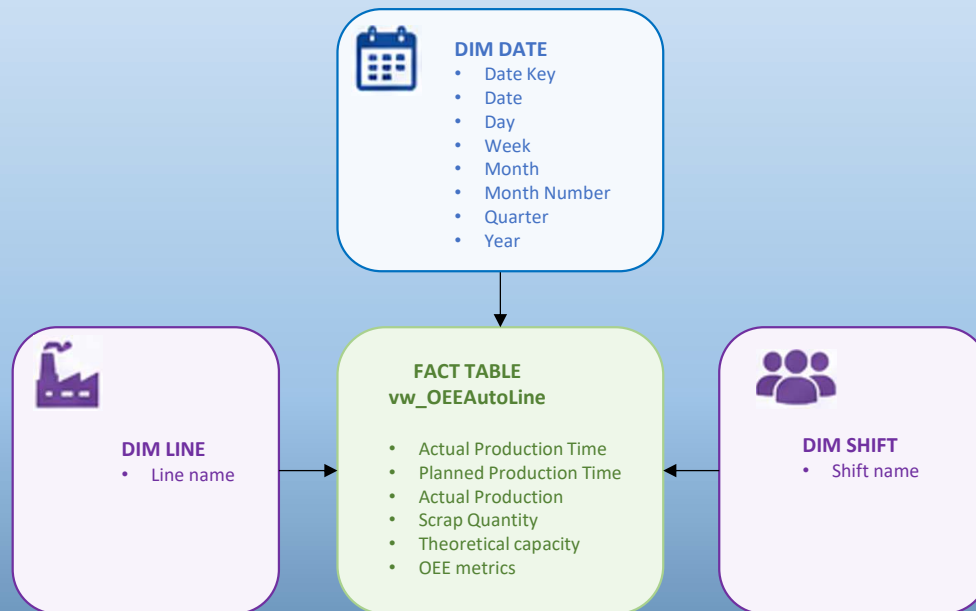


**Efficient
Reporting**



STAR SCHEMA DIAGRAM

Power BI Data Model



KEY BENEFITS



Simple and intuitive data model



High Performance and scalability



Easy to maintain & extend



KPI FRAMEWORK

OOE Calculation Hierarchy

INDUSTRIAL LOAD



$$\frac{\text{Available positions}}{\text{Total positions}}$$

x 100

X

VOLUME LOAD



$$\frac{\text{Planned Positions}}{\text{Available positions}}$$

x 100

X

PERFORMANCE



$$\frac{\text{Actual Production}}{\text{Planned Production}}$$

x 100

X

QUALITY



$$\frac{\text{Good Production}}{\text{Total Production}}$$

x 100

X

AVAILABILITY



$$\frac{\text{Actual Production Time}}{\text{Planned Production Time}}$$

x 100

=

OOE



$$\text{Industrial Load x} \\ \text{Volume Load x} \\ \text{Availability x} \\ \text{Performance x} \\ \text{Quality} \\ \text{x 100}$$



TARGET OOE: 85%

< 60%

Needs
Improvement

60% - 85%

Average
Performance

>85%

World Class
Performance



BUSINESS PROCESS FLOW

From Data to Decision



DATA COLLECTION

Production, Parameters,
Planning and Quality
data captured from
source systems



DATA PROCESSING

Data cleansing,
validation and
transformation in
SQL server



DATA MODELING

Star Schema
modeling for
analysis and
reporting



ANALYSIS & VISUALIZATION

KPIs, dashboards
and drill-down
analysis in Power BI



DECISION MAKING

Actionable insights
support operational and
strategic decisions



BUSINESS VALUE

- ✓ Better Visibility
- ✓ Higher Efficiency
- ✓ Faster Decisions

- ✓ Reduced Losses
- ✓ Continuous Improvement
- ✓ Fast Reactivity



RESPOSITORY ARCHITECTURE

Organized and Scalable Project Structure



01-Automatic_Line_Production_Analytics



datasets

Raw data samples and reference files



documentation

Business logic, data model, and project docs



powerbi

Power BI reports and related files



sql

SQL scripts, views, functions, procedures



screenshots

Dashboards, model and architecture images



README.md

Project overview and documentation



LICENCE

MIT licence



.gitignore

Git ignore file

BENEFITS



Well Organized

Clear folder structure
For easy navigation



Scalable

Easy to extend and
maintain



Reproducible

All scripts and docs
version controlled



Professional

Industry standard
project layout